

AI+RES: forecasting rare weather extremes three weeks out, at a fraction of the compute

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The result in five lines.

- Three weeks ahead, a standard 24-run forecast ensemble does not produce a genuinely extreme week. The event is rarer than 1-in-24, so the ensemble reports zero and you learn nothing.
- Our method runs 64 forecasts and, five times along the way, kills the ones going nowhere and clones the ones heading toward the event. A second, independent AI weather model does the picking.
- On the seven events at or beyond the 96th percentile of climate history, it put forecast members on the observed event **6 times out of 7**. Standard ensemble sampling: **0 of 7**. The U.S. operational model (CFSv2): **0 of 7**.
- **One forecast costs about \$155 and takes 7 to 9 hours** on a single machine with eight H100 GPUs.
- Getting the same set of extreme trajectories by brute force is estimated at **5x to 23x the compute cost**, and the multiple grows the rarer the event is.

This is a **qualitative** assessment. Nine events is enough to show the method reaches events the alternatives miss; it is not enough to state a hit rate, a skill score, or a calibrated probability with statistical confidence. Every number below is a measurement from a real run unless marked *estimated*.

1. The problem

Weather forecasts are run as ensembles: the same model is launched many times from slightly different starting points, and the spread of the outcomes is the forecast. The number of runs sets the smallest probability the ensemble can see. With 24 runs you cannot resolve anything rarer than about 1-in-24, and a single run landing on the event is indistinguishable from luck.

Three weeks out, the events that matter most are exactly the ones in that blind spot. The 2021 Pacific Northwest heat dome ran 7.72 K above normal over its region for a week, an event without precedent in 64 years of record for that place and season. A 24-member state-of-the-art AI ensemble launched three weeks earlier produced zero runs that reached it. The operational U.S. government model, CFSv2, forecast a week essentially at normal. Neither model is wrong about the atmosphere; both are being asked a question their sample size cannot answer.

2. What we built

Rare-event sampling spends the compute budget where the answer is, instead of spreading it evenly. We launch 64 forecast trajectories from the same starting conditions. Every three days we stop them, score each one, discard the low scorers, and duplicate the high scorers in their place, so the population always has 64 members but is progressively concentrated on the trajectories heading toward the extreme. Bookkeeping weights carried alongside each trajectory convert the biased population back into an unbiased probability at the end.

The step that makes this work is the score. The traditional choice is “how hot is this trajectory right now,” which three weeks before a heat wave carries no information about which trajectory becomes hot. We instead score each trajectory by running a second, independently trained AI weather model forward from it and asking what it

forecasts for the target week. We measured the difference: with the AI score, the ranking at the 9-day checkpoint correlates 0.80 with where trajectories actually ended up; with the traditional score it correlates 0.06. Run on the identical setup with only the score swapped, the traditional method reaches the 2021 heat dome with 0 of 64 trajectories.

Two components: the trajectory model is GenCast (Google DeepMind) at 0.25-degree resolution; the scoring model is FourCastNet 3 (NVIDIA). Both were trained on data ending before 2019, so every event tested here is out of sample – the models had never seen it.

3. How it performs

Table 1: Nine test cases, all forecast from three weeks before the event. A_L is the event’s temperature anomaly: how far the week-long regional average ran from normal, in kelvin (1 K = 1.8 °F). Each method column is its *most extreme member* – the best shot that method took at the event. **Bold** marks a method that reached or passed what was observed. Percentile is the event’s rarity against 1959–2022 reanalysis for that region and season, with the estimated return period in parentheses.

		A_L , week-mean temperature anomaly (K)			
event	percentile	observed	AI+RES	CFSv2	direct sampling
<i>Genuine extremes</i>					
S. Texas heat dome, Jun 2023	>99.96 (record)	+5.73	+6.19	+1.19	+4.19
Southwest heat wave, Aug 2020	99.96 (1-in-210 y)	+5.14	+4.31	+1.60	+3.82
California heat wave, Sep 2022	99.71 (1-in-37 y)	+4.71	+6.33	+3.68	+4.70
Pacific NW heat dome, Jun 2021	99.60 (1-in-22 y)	+7.72	+10.29	+3.42	+7.08
CONUS warm week, Dec 2025	98.78 (1-in-10 y)	+3.26	+5.12	+0.22	+3.46
Winter Storm Elliott, Dec 2022	98.41 (1-in-5 y)	−14.42	−17.04	−10.74	−9.83
Winter Storm Uri, Feb 2021	96.80 (1-in-4 y)	−7.40	−10.20	−3.86	−6.51
<i>Controls, deliberately mild</i>					
CONUS warm week, Aug 2024	94.08 (1-in-4 y)	+1.06	+2.37	+1.35	+1.96
CONUS median week, Nov 2023	60.10 (typical)	+0.21	+4.16	+1.75	+3.14
reached, 7 genuine extremes			6 of 7	0 of 7	0 of 7
reached, all 9			8 of 9	2 of 9	3 of 9

Lead times, which are not identical across columns. AI+RES forecasts from a 21-day lead. CFSv2 is also at 21 days, as four operational cycles launched 6 hours apart. Direct sampling is a 24-member GenCast ensemble at a 14-day lead in every row except the S. Texas case, where it is 16 GenCast runs at 21 days; a separate 24-member FourCastNet 3 ensemble at 21 days was also run on five of the cases. The direct-sampling column therefore had, in most rows, a one-week head start on AI+RES, and still did not reach any of the seven extremes. The comparison is conservative in the direct column’s favor.

What the table says. Where the event is genuinely rare, AI+RES puts members on it and the alternatives do not. Where the event is ordinary – the two controls – all three methods reach it, which is the check that the difference is about rarity and not about one model being better in general. The one miss, the 2020 Southwest heat wave, is the rarest case in the set by return period; the method moved the population 1.75 K toward it and stopped 4.4 K short.

The qualitative bound on what this setup can do. On this evidence, the configuration tested here (64 trajectories, five resampling steps, 21-day lead) reliably produces the event for extremes up to roughly the **99.9th percentile** – about a 1-in-40-year week for a given region and season. It reached one event past the 64-year record and failed on another, so the boundary is near, not beyond, the 1-in-100-year level for this budget. Past that, the limit is population size and is bought back linearly: more trajectories, proportionally more compute. This is a read on nine cases, not a measured hit rate.

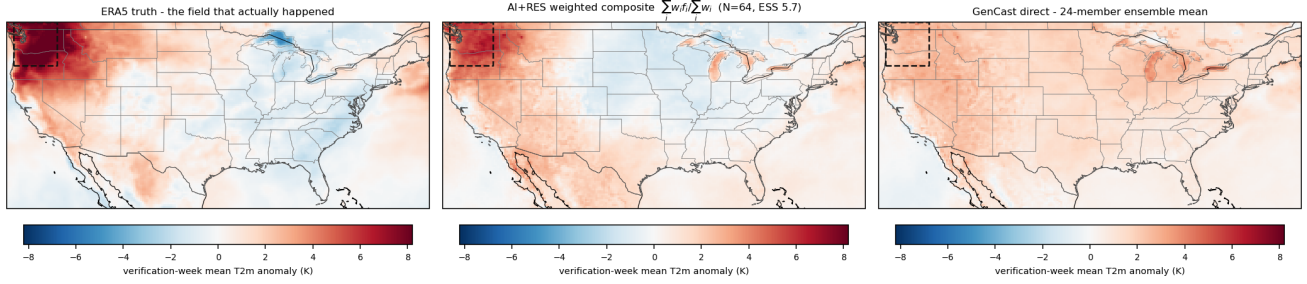


Figure 1: The 2021 Pacific Northwest heat dome, as maps of the week-average temperature anomaly. **Left:** what happened. **Middle:** the AI+RES population three weeks out – the dome is in the right place with the right intensity. **Right:** the standard 24-member ensemble average from the same starting point, which contains no dome at all. Dashed box is the scored region. The middle panel is a weighted composite of trajectories selected for this event, so it is a picture of where the method put its samples, not a forecast in the ordinary sense.

4. Cost and speed

Table 2: One AI+RES forecast, measured, on one node of eight NVIDIA H100 GPUs. Dollar figures use \$2.50 per GPU-hour (\$20/hour for the node), a mid-market on-demand cloud rate.

hardware	1 node, 8x NVIDIA H100 (80 GB)
settings	64 trajectories, 5 resampling steps every 3 days, 6 scoring runs per trajectory
forecast horizon	21 days, scored over the final 7-day window
wall-clock, one forecast	7.2 to 8.7 hours
compute, one forecast	61 to 64 H100-hours
cost, one forecast	\$153 to \$160
throughput	roughly 3 forecasts per node per day

The brute-force alternative. The other way to get trajectories that reach the event is to run an enormous plain ensemble until some of its members land there by chance. Table 3 prices that. It takes the probability AI+RES estimated for each event, asks how many plain forecast runs would be needed to expect the same number of members on the event, and multiplies by the measured cost of one 21-day run (about 0.4 H100-hours, roughly \$1.00).

Table 3: *Estimated, not measured.* Brute-force cost to obtain the same tail sample AI+RES delivered, against the \$156 AI+RES run. Derived from the AI+RES probability estimate for each event and a measured per-run cost; no ensemble of this size was actually run, and the probability estimates themselves carry roughly a factor-of-two uncertainty at 64 trajectories.

event	members AI+RES put on target	event probability	plain runs needed (est.)	brute-force cost (est.)	cost multiple
Pacific NW heat dome, Jun 2021	42 of 64	1-in-19	780	\$780	5x
California heat wave, Sep 2022	25 of 64	1-in-53	1,320	\$1,320	8x
S. Texas heat dome, Jun 2023	6 of 64	1-in-370	2,220	\$2,220	14x
Winter Storm Elliott, Dec 2022	20 of 64	1-in-127	2,530	\$2,530	16x
Winter Storm Uri, Feb 2021	32 of 64	1-in-111	3,560	\$3,560	23x

Three things to take from Table 3. First, the saving is real but it is not the headline – the headline is that at the \$156 budget, brute force returns *nothing at all* on these events. That is the measured result in Table 1: 0 of 7. Second, the multiple grows with rarity, from 5x on the least rare event to 23x on the rarest one priced here, because brute-force cost scales as one over the event probability while AI+RES cost is flat. Third, the 2020 Southwest heat wave is absent from Table 3 because neither method reached it; brute force would need an estimated 3,300+ runs (\$3,300+) for a *single* member, and AI+RES at this budget did not get there either.

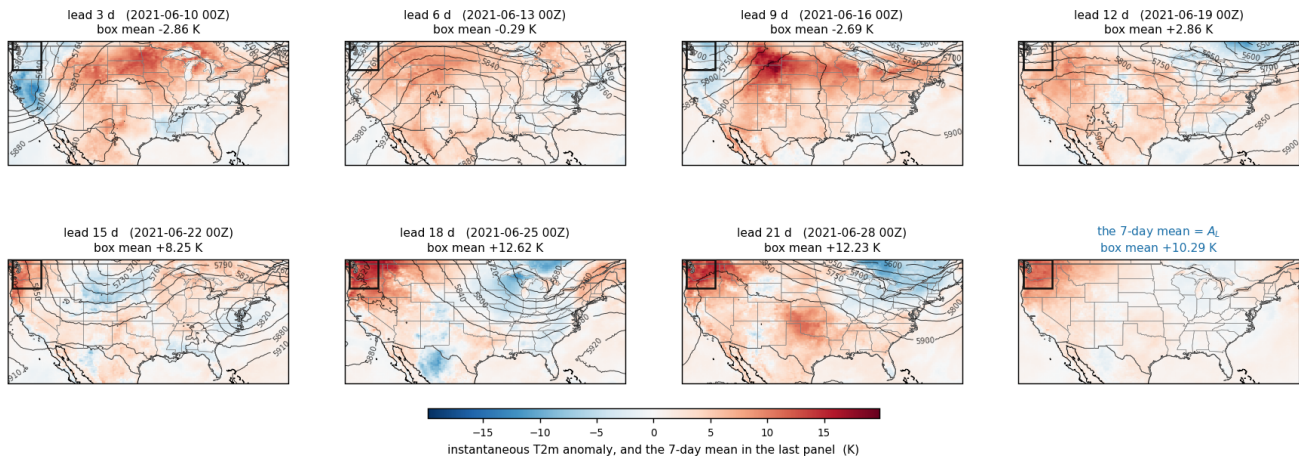


Figure 2: Why the AI score is the ingredient. This is the single hottest surviving trajectory, followed from the 3-day checkpoint to the event. At the 9-day checkpoint it had the *fourth-coolest* surface temperature of the 64 trajectories alive and would have been near the bottom of any “how hot is it now” ranking. The scoring model ranked it tenth and gave it the maximum eleven clones, because the upper-atmosphere ridge (contours) that later drives the heat dome was already building offshore while the surface beneath it had not yet warmed.

5. What this is, and what it is not

It is a demonstration, on nine out-of-sample cases, that concentrating a fixed compute budget on the trajectories heading toward an extreme produces forecast members on events that conventional sampling misses entirely at the same or larger budget, at a measured \$155 and 8 hours per forecast.

It is not a calibrated probability service. Three limits, stated plainly:

1. **Nine events cannot establish a hit rate.** The 6-of-7 figure has a wide interval around it. A 42-case hindcast campaign designed to produce a proper reliability curve is built and staged; it has not been run.
2. **The probabilities carry about a factor of two.** They are internally consistent – where plain sampling can resolve a probability at all, our estimate agrees with it – but the absolute level at 64 trajectories is noisy, and the method gives no probability at all for an event milder than the population it has already concentrated.
3. **It samples the AI model’s forecast distribution, not the atmosphere’s.** The claim supported is that the method reaches the extreme tail of a state-of-the-art forecast model efficiently. Whether that model’s tail is itself unbiased against the real climate is a separate question this work does not settle.

Next. The 42-case calibration campaign, at an estimated \$1,700 of GPU time for its first 12-case tier; a scoring function for hurricanes, where the weekly regional average used here washes out a one-day landfall and a different target is needed; and lighter model pairs, since the current pair requires an 80 GB GPU per trajectory and a smaller one would cut the per-forecast cost directly.